

CSE 220 SIGNALS AND SYSTEMS · FINAL EVALUATION

Sonar Distance Simulator

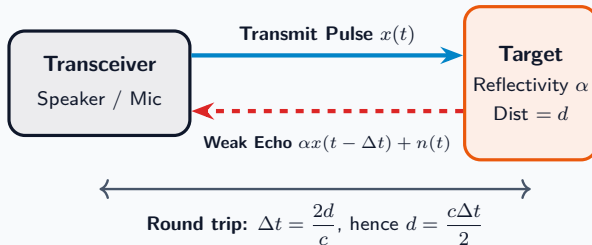
Pulse-Echo Ranging, Matched Filtering, and Range Resolution Limits

Team SevenEight

Aditya Tirtho Roy (2305178) | **Farhan Ehsas Sami** (2305177)

Department of Computer Science and Engineering, BUET

Pure DSP Core · Monte Carlo Analysis · Streamlit Instrument UI · Real Hardware Ranging



1. Severe Noise

Echoes weaken as $1/r^2$. At 0 dB SNR, the raw waveform can hide them completely.

2. Echo Merging

Closely spaced targets create overlapping echoes that appear as one merged return.

3. Energy Dilemma

Short pulses resolve targets but carry little energy; long pulses are stronger but blur targets.

1. INPUTS

Pulse Controls:

- Waveform: Rect, Gauss, Chirp
- Duration T (e.g., 2 ms)
- Carrier f_c & Bandwidth B

Acoustic Channel:

- True Distances d_1, d_2
- Noise Level SNR_{dB}
- Speed $c = 343$ m/s

Configure

2. DSP ENGINE

Acoustic Channel

- Delay:
 $k = \text{round}(2df_s/c)$
- AWGN Noise Injection

Matched Filter

- $$r_{yx}[k] = \sum y[n]x[n-k]$$

Hilbert Envelope

- $e[n] = |\tilde{r}[n]|$, Peak Finder

Results

3. OUTPUTS

Ranging Numbers:

- Estimated Distance $\hat{d}$ (m)
- Absolute Error $|d - \hat{d}|$
- Multi-target list $[\hat{d}_1, \hat{d}_2]$

Visual Plots:

- Noisy Audio Buffer
- Sharp Correlation Spike

Audible Sound:

- Listenable .wav pings

Core Takeaway: Configure the pulse and scene, then receive distance estimates, diagnostic plots, and audible sonar clips.

Sliding Template Cross-Correlation:

- Mathematically optimal filter in AWGN (Cauchy-Schwarz Inequality):

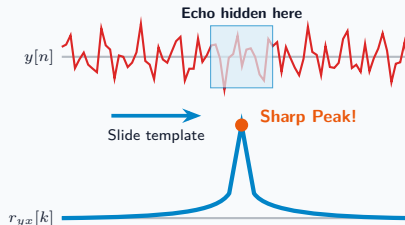
$$r_{yx}[k] = \sum_n y[n] x[n - k]$$

- Intuition:**

- Where there is only noise $\rightarrow$ multiplies random numbers $\rightarrow$ **averages to zero**.
- Where the echo matches the template $\rightarrow$ multiplies in-phase $\rightarrow$ **constructive surge!**

- Peak to Distance:**

$$\hat{d} = \frac{(k_{\text{peak}} - \text{offset}) \cdot c}{2f_s}$$



The Aha! Moment: Even when noise is louder than the echo (0 dB SNR), the matched filter recovers the echo with sub-millimeter precision!

1. The Carrier Trap (Raw Correlation):

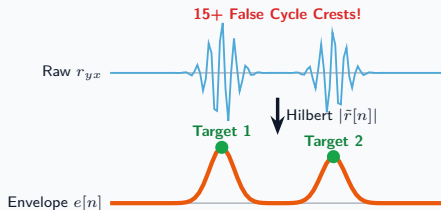
- Carrier oscillates at 4 kHz.
- Raw correlation produces **15–20 cycle ripples** for just 2 true targets!
- Simple thresholding counts false targets.

2. The Solution (Hilbert Envelope):

- Construct analytic signal:

$$\tilde{r}[n] = r[n] + j\mathcal{H}\{r[n]\}$$

- Smooth envelope: $e[n] = |\tilde{r}[n]|$.
- Peak finder on $e[n]$ detects **exactly 2 true target crests!**



Rule of Thumb: Raw argmax for single targets (sharpest); Hilbert envelope for multi-targets (eliminates carrier cycle ripples).

The Rayleigh Resolution Limit:

$$\Delta R_{\min} \approx \frac{c}{2B}$$

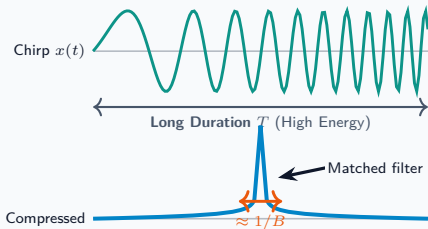
Two targets can only be separated if their echo gap exceeds the pulse's correlation width.

The Classical Trade-Off (Tone Bursts):

- Bandwidth is tied to duration: $B \approx 1/T$.
- To separate close targets $\rightarrow$ must make pulse tiny $\rightarrow$ carries tiny energy $\rightarrow$ ruined by noise!

The Chirp (LFM) Breakthrough:

- Sweeps frequency over time ($f_0 \rightarrow f_0 + B$).
- **Decouples duration T from bandwidth B :**
high energy + razor-sharp resolution!



Pulse Compression: Long duration plus wide bandwidth gives high energy and a narrow $\approx 1/B$ correlation peak.

The Golden Rule

Functions in `src/` take numbers/arrays and return numbers/arrays. They NEVER call `plt.show()` or `print()`.

Why This Matters:

- **100% Reusable DSP Core:** Tests, report figures, and the Streamlit web app call the **exact same functions**.
- **Zero UI Coupling:** Front-end UI redesigns never risk breaking core physics calculations.
- **Fully Automated:** 33 passing pytest tests + 17 pulse physics verification checks.

`src/pulse.py` · Rect / Gaussian / Chirp generators; FFT -3 dB bandwidth



`src/channel.py` · Sample delay, geometric loss, AWGN injection



`src/receiver.py` · Matched filter, single-target & Hilbert multi-target

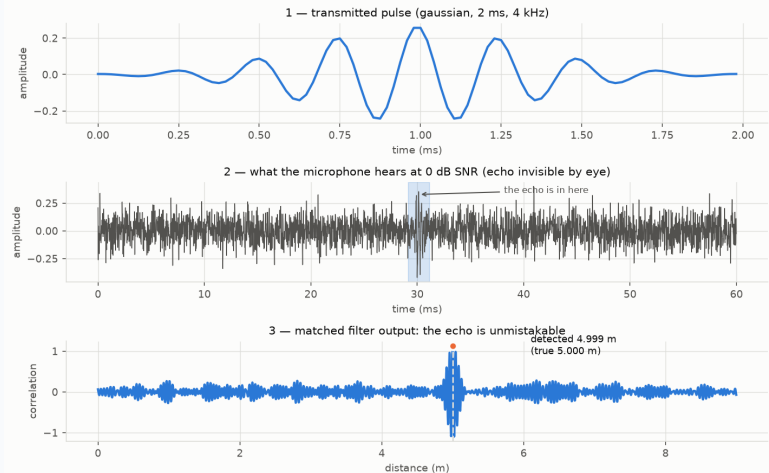


`src/evaluate.py` · Monte Carlo noise & resolution sweeps



Shared Consumers

tests/ (33 tests) | notebooks/
| app/streamlit_app.py



1. Transmit:

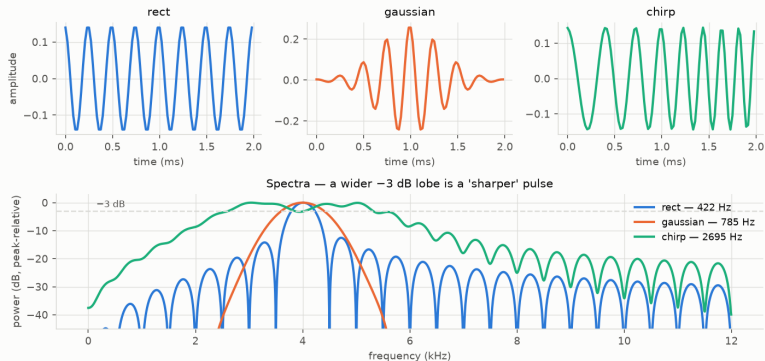
Gaussian pulse (2 ms, 4 kHz) normalized to unit energy.

2. Corrupted:

At 0 dB SNR, echo is completely drowned by ambient noise!

3. Recovered:

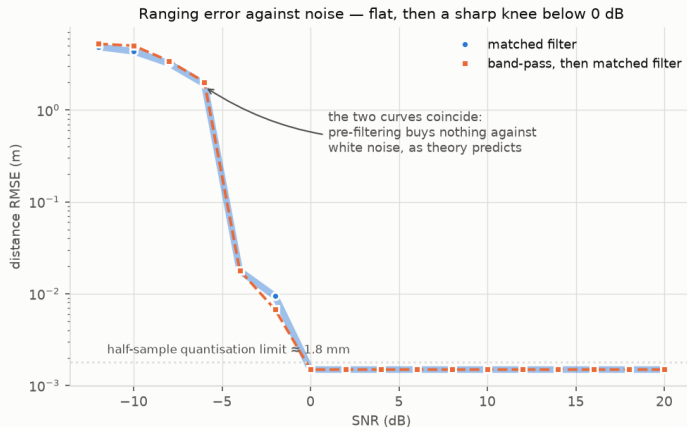
Matched filter yields a towering peak at 5.000 m (< 1 mm error!).



Waveform Comparison:

- **Rect:** Abrupt on/off switch → heavy spectral splatter.
- **Gaussian:** Bell envelope → smooth, clean spectrum.
- **Chirp:** Linear frequency sweep → wide -3 dB bandwidth for sharp target separation.

Key Metric: A wider -3 dB spectral lobe yields a sharper correlation peak.



Key Observations:

1. Quantization Floor (≈ 1.8 mm):

- Above 0 dB, error is flat.
- Arises from discrete sampling:

$$\text{Floor} \approx \frac{c}{4f_s} \approx 1.79 \text{ mm}$$

2. The Threshold Cliff:

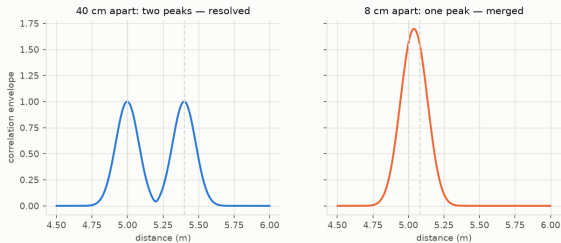
- Below -4 dB, noise spikes eclipse the echo peak, causing rapid breakdown.

3. Bandpass Filter Overlap:

- Pre-filtering coincides with raw matched filter $\rightarrow$ confirms matched filter is already optimal for white noise!

Multipath Separation (40 cm vs. 8 cm)

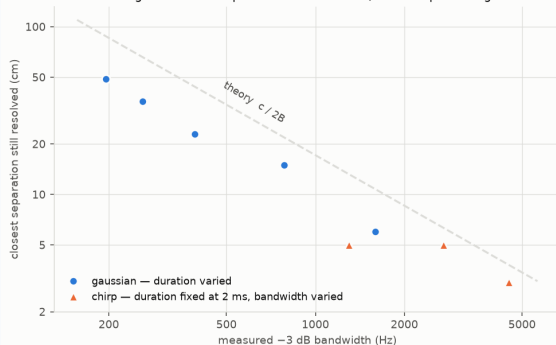
Two targets, same pulse — the limit is about 15 cm here



- 40 cm gap: Two distinct peaks resolved.
- 8 cm gap: Echoes merge into a single wide blob.

Measured Resolution vs. Bandwidth

Range resolution improves with bandwidth, not with pulse length



- Measured points track theoretical dashed curve $c/(2B)$ across a 23× bandwidth range.
- Chirps fix duration at 2 ms while sweeping bandwidth.